THESIS_PRIMARY_regression_equal_vs_shrinkage.csv

method	contrast	estimate	lower_Shrinkage	lower_Equal−variance	upper_Shrinkage	upper_Equal−variance	width_Shrinkage	width_Equal−variance
Regression MI	1 vs 0	62.255	31.151	31.038	93.359	93.472	62.208	62.434
Regression MI	2 vs 0	70.308	41.464	41.334	99.152	99.281	57.688	57.947
Regression MI	3 vs 0	49.013	20.455	20.404	77.571	77.622	57.116	57.218

THESIS_PRIMARY_regression_equal_vs_shrinkage.csv

significant_Shrinkage	significant_Equal−variance
TRUE	TRUE
TRUE	TRUE
TRUE	TRUE

REFERENCE_METHOD_propensity_equal_vs_shrinkage.csv

method	contrast	estimate	lower_Shrinkage	lower_Equal−variance	upper_Shrinkage	upper_Equal−variance	width_Shrinkage	width_Equal−variance
Propensity MI	1 vs 0	59.540	29.715	29.500	89.365	89.581	59.650	60.081
Propensity MI	2 vs 0	69.088	38.642	38.418	99.535	99.758	60.892	61.340
Propensity MI	3 vs 0	50.353	20.821	20.807	79.885	79.899	59.064	59.092

REFERENCE_METHOD_propensity_equal_vs_shrinkage.csv

significant_Shrinkage	significant_Equal−variance
TRUE	TRUE
TRUE	TRUE
TRUE	TRUE

ACTG175_SUPPLEMENTARY_PRIMARY_RESULT_TABLE.csv

Method	Contrast	Estimate	Shrinkage 95% CI	Non−shrinkage 95% CI	Width Sh	Width NS	Conclusion (Sh / NS)
Complete−case	1 vs 0	53.635	[21.859, 85.411]	[22.348, 84.923]	63.552	62.575	Significant / Significant
Complete−case	2 vs 0	67.202	[35.519, 98.885]	[36.030, 98.375]	63.366	62.345	Significant / Significant
Complete−case	3 vs 0	41.175	[9.802, 72.549]	[9.884, 72.466]	62.747	62.582	Significant / Significant
Regression MI	1 vs 0	62.255	[31.151, 93.359]	[31.334, 93.176]	62.208	61.843	Significant / Significant
Regression MI	2 vs 0	70.308	[41.464, 99.152]	[41.712, 98.903]	57.688	57.191	Significant / Significant
Regression MI	3 vs 0	49.013	[20.455, 77.571]	[20.530, 77.495]	57.116	56.965	Significant / Significant
PMM MI	1 vs 0	61.237	[31.370, 91.104]	[31.508, 90.966]	59.734	59.458	Significant / Significant
PMM MI	2 vs 0	69.903	[41.060, 98.746]	[41.359, 98.447]	57.685	57.088	Significant / Significant
PMM MI	3 vs 0	51.746	[21.176, 82.316]	[21.138, 82.355]	61.141	61.217	Significant / Significant
Propensity MI	1 vs 0	59.540	[29.715, 89.365]	[29.933, 89.147]	59.650	59.214	Significant / Significant
Propensity MI	2 vs 0	69.088	[38.642, 99.535]	[38.837, 99.339]	60.892	60.502	Significant / Significant
Propensity MI	3 vs 0	50.353	[20.821, 79.885]	[20.852, 79.853]	59.064	59.001	Significant / Significant

APPENDIX_group_and_missingness_summary.csv

group	n_total	n_observed	n_missing	missing_pct	observed_mean	observed_sd	observed_variance
0	532	321	211	39.662	287.617	166.383	27683.41
1	522	333	189	36.207	341.252	173.585	30131.86
2	524	337	187	35.687	354.819	173.369	30056.80
3	561	351	210	37.433	328.792	178.255	31774.87

APPENDIX_missingness_model_coefficients.csv

term	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	−0.418	0.089	−4.714	0.000
group1	−0.152	0.127	−1.191	0.234
group2	−0.170	0.127	−1.337	0.181
group3	−0.099	0.125	−0.798	0.425
x	−0.104	0.045	−2.278	0.023

APPENDIX_shrinkage_alpha.csv

method	contrast	shrink_alpha
Complete−case	1 vs 0	1.000
Complete−case	2 vs 0	1.000
Complete−case	3 vs 0	1.000
Regression MI	1 vs 0	0.750
Regression MI	2 vs 0	0.750
Regression MI	3 vs 0	0.750
PMM MI	1 vs 0	0.808
PMM MI	2 vs 0	0.808
PMM MI	3 vs 0	0.808
Propensity MI	1 vs 0	0.577
Propensity MI	2 vs 0	0.577
Propensity MI	3 vs 0	0.577

APPENDIX_x_outcome_relation.csv

correlation_cd40_cd496	natural_missing_pct
0.531	37.26

ALL_METHODS_full_precision_intervals.csv

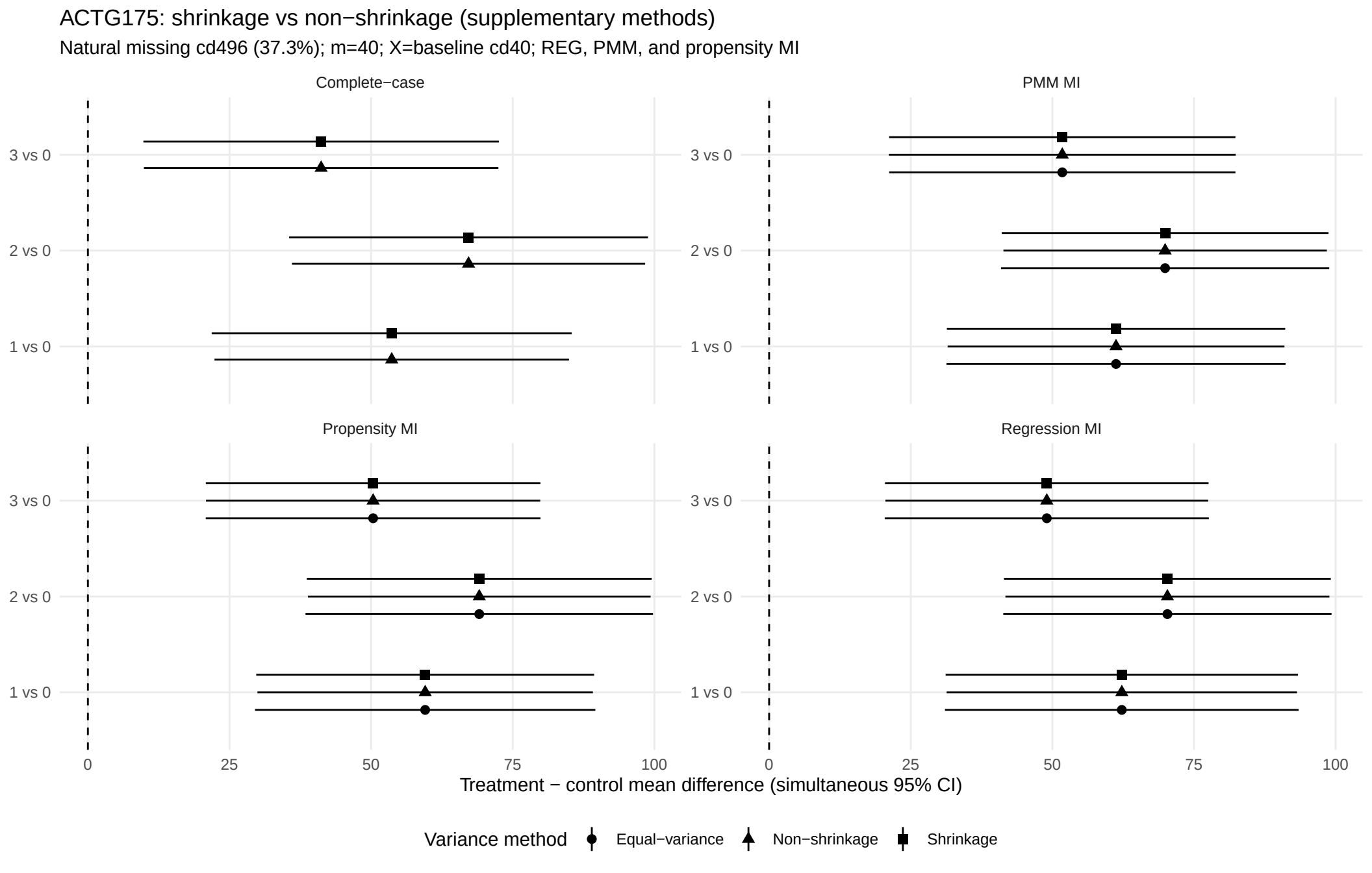
method	contrast	estimate	lower	upper	width	significant	variance_method	shrink_alpha
Complete−case	1 vs 0	53.635	21.859	85.411	63.552	TRUE	Shrinkage	1.000
Complete−case	2 vs 0	67.202	35.519	98.885	63.366	TRUE	Shrinkage	1.000
Complete−case	3 vs 0	41.175	9.802	72.549	62.747	TRUE	Shrinkage	1.000
Complete−case	1 vs 0	53.635	22.348	84.923	62.575	TRUE	Non−shrinkage	NA
Complete−case	2 vs 0	67.202	36.030	98.375	62.345	TRUE	Non−shrinkage	NA
Complete−case	3 vs 0	41.175	9.884	72.466	62.582	TRUE	Non−shrinkage	NA
Regression MI	1 vs 0	62.255	31.151	93.359	62.208	TRUE	Shrinkage	0.750
Regression MI	2 vs 0	70.308	41.464	99.152	57.688	TRUE	Shrinkage	0.750
Regression MI	3 vs 0	49.013	20.455	77.571	57.116	TRUE	Shrinkage	0.750
Regression MI	1 vs 0	62.255	31.334	93.176	61.843	TRUE	Non−shrinkage	NA
Regression MI	2 vs 0	70.308	41.712	98.903	57.191	TRUE	Non−shrinkage	NA
Regression MI	3 vs 0	49.013	20.530	77.495	56.965	TRUE	Non−shrinkage	NA
Regression MI	1 vs 0	62.255	31.038	93.472	62.434	TRUE	Equal−variance	NA
Regression MI	2 vs 0	70.308	41.334	99.281	57.947	TRUE	Equal−variance	NA
Regression MI	3 vs 0	49.013	20.404	77.622	57.218	TRUE	Equal−variance	NA
PMM MI	1 vs 0	61.237	31.370	91.104	59.734	TRUE	Shrinkage	0.808
PMM MI	2 vs 0	69.903	41.060	98.746	57.685	TRUE	Shrinkage	0.808
PMM MI	3 vs 0	51.746	21.176	82.316	61.141	TRUE	Shrinkage	0.808
PMM MI	1 vs 0	61.237	31.508	90.966	59.458	TRUE	Non−shrinkage	NA
PMM MI	2 vs 0	69.903	41.359	98.447	57.088	TRUE	Non−shrinkage	NA

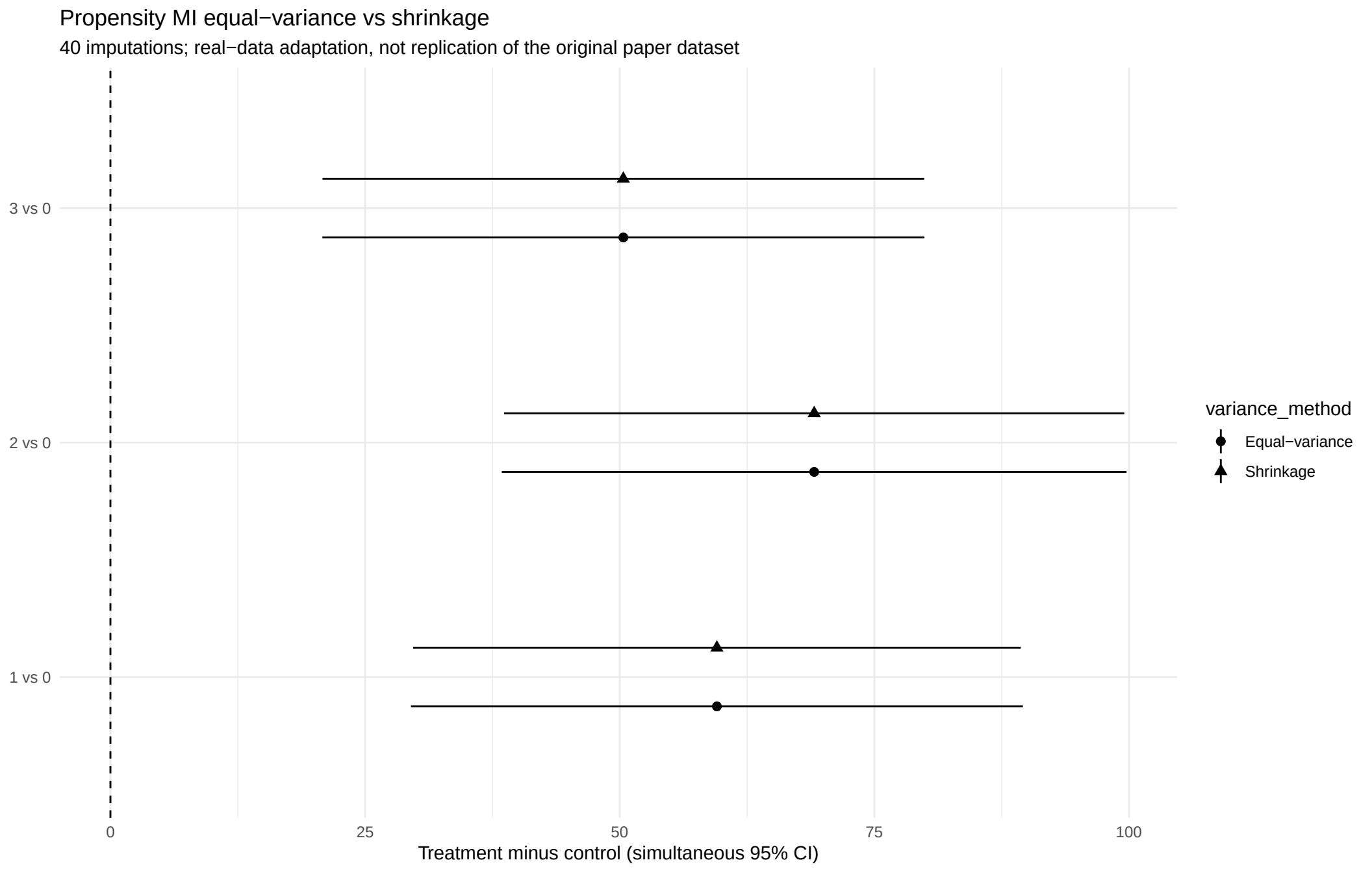
ALL_METHODS_full_precision_intervals.csv

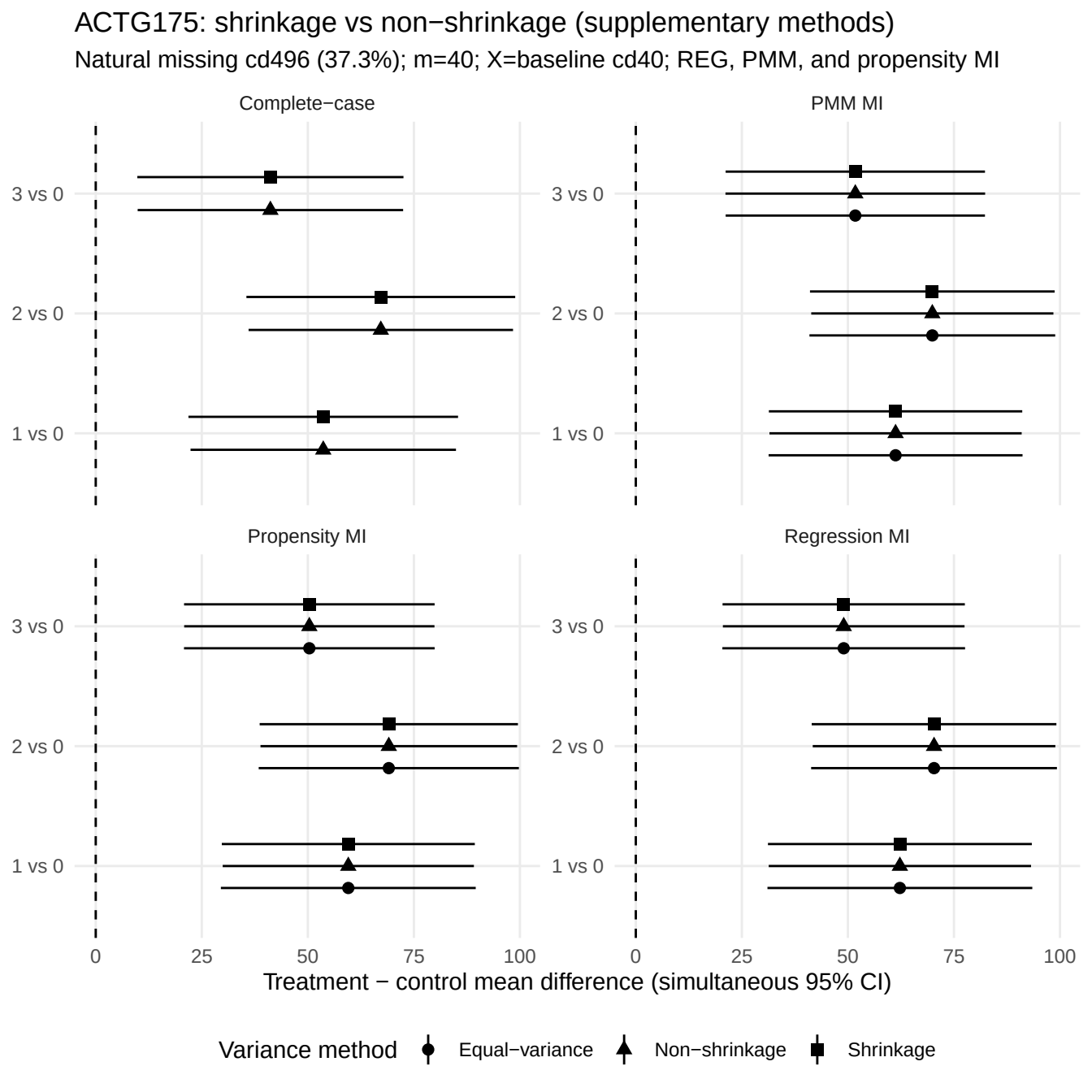
method	contrast	estimate	lower	upper	width	significant	variance_method	shrink_alpha
PMM MI	3 vs 0	51.746	21.138	82.355	61.217	TRUE	Non−shrinkage	NA
PMM MI	1 vs 0	61.237	31.307	91.167	59.860	TRUE	Equal−variance	NA
PMM MI	2 vs 0	69.903	40.935	98.871	57.937	TRUE	Equal−variance	NA
PMM MI	3 vs 0	51.746	21.182	82.310	61.128	TRUE	Equal−variance	NA
Propensity MI	1 vs 0	59.540	29.715	89.365	59.650	TRUE	Shrinkage	0.577
Propensity MI	2 vs 0	69.088	38.642	99.535	60.892	TRUE	Shrinkage	0.577
Propensity MI	3 vs 0	50.353	20.821	79.885	59.064	TRUE	Shrinkage	0.577
Propensity MI	1 vs 0	59.540	29.933	89.147	59.214	TRUE	Non−shrinkage	NA
Propensity MI	2 vs 0	69.088	38.837	99.339	60.502	TRUE	Non−shrinkage	NA
Propensity MI	3 vs 0	50.353	20.852	79.853	59.001	TRUE	Non−shrinkage	NA
Propensity MI	1 vs 0	59.540	29.500	89.581	60.081	TRUE	Equal−variance	NA
Propensity MI	2 vs 0	69.088	38.418	99.758	61.340	TRUE	Equal−variance	NA
Propensity MI	3 vs 0	50.353	20.807	79.899	59.092	TRUE	Equal−variance	NA

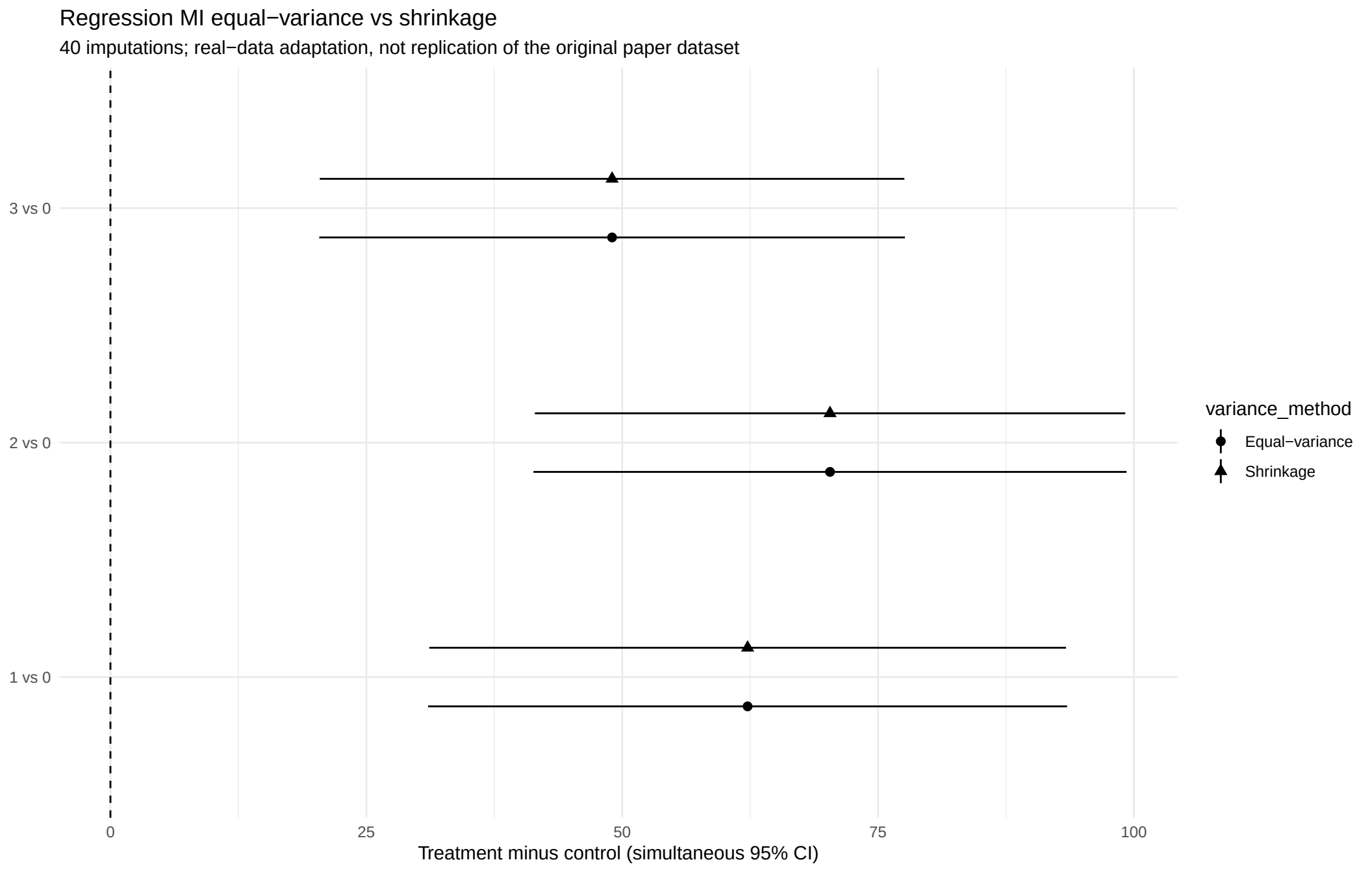
completion_summary.csv

analysis	imputations
checkpoints/pmm_20461107	40
checkpoints/propensity_20561107	40
checkpoints/regression_20361107	40


ACTG175: shrinkage vs non−shrinkage (supplementary methods)
Natural missing cd496 (37.3%); m=40; X=baseline cd40; REG, PMM, and propensity MI
Complete−case
PMM MI
Propensity MI
Regression MI
3 vs 0
2 vs 0
1 vs 0
0
25
50
75
100
Treatment − control mean difference (simultaneous 95% CI)
Variance method
Equal−variance
Non−shrinkage
Shrinkage


Propensity MI equal−variance vs shrinkage
40 imputations; real−data adaptation, not replication of the original paper dataset
3 vs 0
2 vs 0
1 vs 0
0
25
50
75
100
Treatment minus control (simultaneous 95% CI)
variance_method
Equal−variance
Shrinkage


ACTG175: shrinkage vs non−shrinkage (supplementary methods)
Natural missing cd496 (37.3%); m=40; X=baseline cd40; REG, PMM, and propensity MI
Complete−case
PMM MI
Propensity MI
Regression MI
3 vs 0
2 vs 0
1 vs 0
0
25
50
75
100
Treatment − control mean difference (simultaneous 95% CI)
Variance method
Equal−variance
Non−shrinkage
Shrinkage


Regression MI equal−variance vs shrinkage
40 imputations; real−data adaptation, not replication of the original paper dataset
3 vs 0
2 vs 0
1 vs 0
0
25
50
75
100
Treatment minus control (simultaneous 95% CI)
variance_method
Equal−variance
Shrinkage